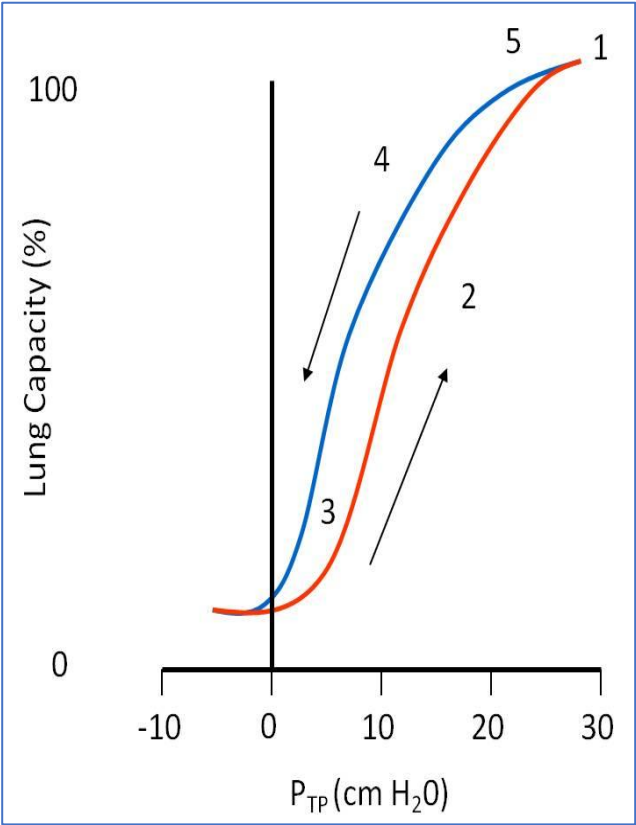


CPR IMCQ 7 Fall 2025

10 Physiology Questions

1. A 45-year-old woman comes to the physician for a routine examination. She has no history of any illness. Vital signs are within normal limits. The pressure-volume relationship of the lung of this patient during one breathing cycle is shown in the graph. Which of the following sets of lung features is best represented at point 1 in this graph?

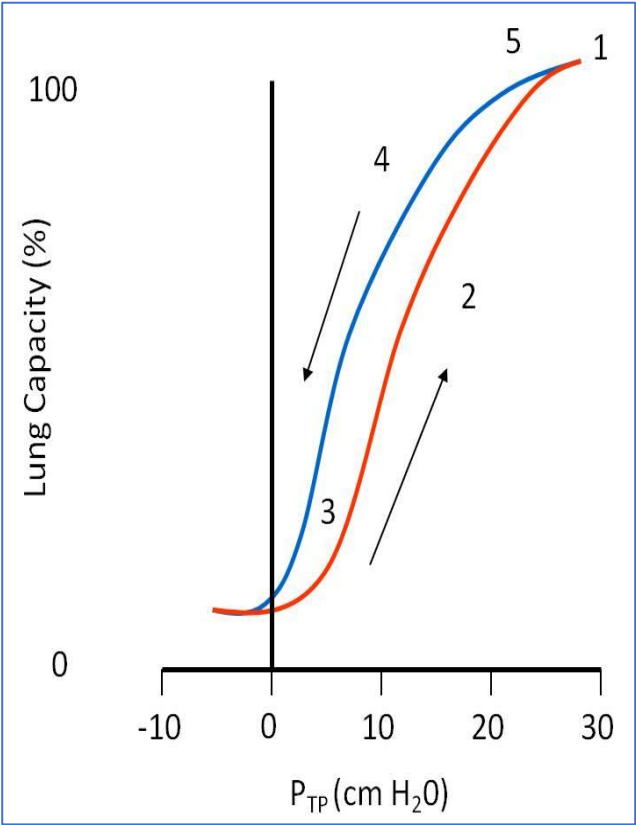
Option	Lung Recoil	Alveolar size	Distribution of surfactants in	Surface Tension in alveoli
A	Low	Large	Smaller surface area	High
B	High	Large	Larger surface area	High
C	High	Small	Larger surface area	Low
D	Low	Small	Smaller surface area	High
E	High	Large	Larger surface area	Low



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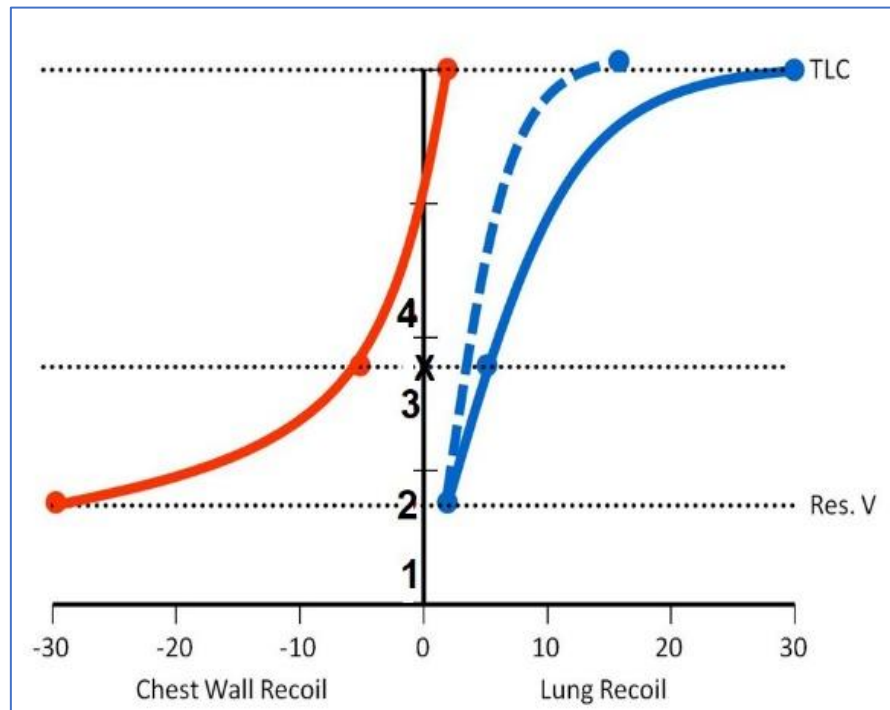
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E	High	Large	Larger surface area	Low

SOM.MK.II.BPM1.3.CPR.2.PHYS.0204: Explain surface tension and how it applies to lung mechanics, including the effects of alveolar size and the role of surfactants. Infer the role of surfactants in preventing atelectasis.



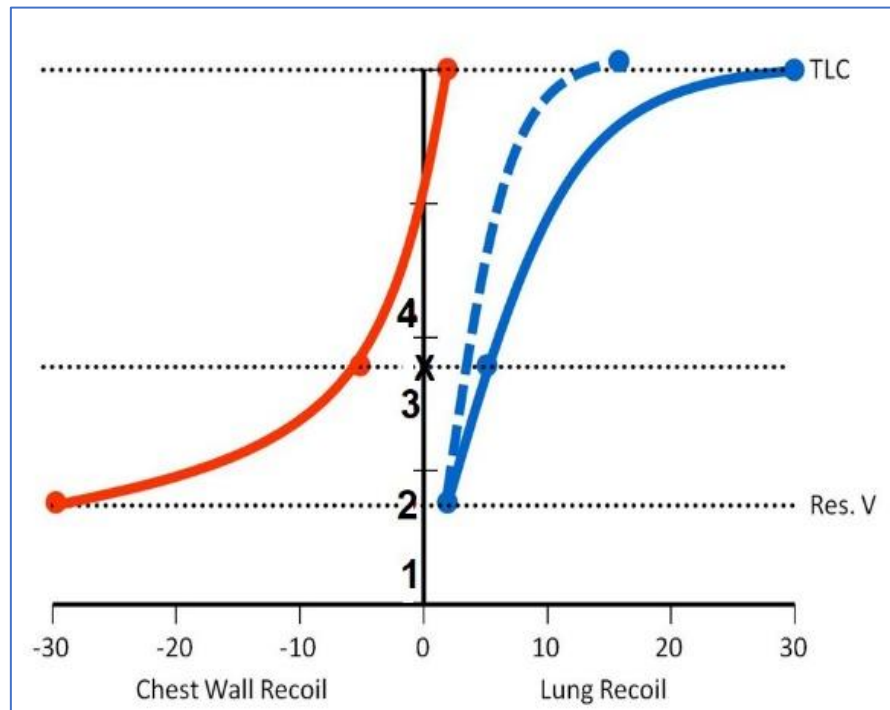
2. A 54-year-old man comes to the physician because of a 6-month history of progressive shortness of breath. He has smoked 2 packs of cigarettes daily for 40 years. Physical examination shows increased use of accessory muscles during expiration. The lung compliance of the patient is shown as blue discontinuous curve in the graph. Which of the following best represents this patient's FRC and lung recoil ability?

- A. 1, high lung recoil
- B. 2, high lung recoil
- C. 3, high lung recoil
- D. 4, high lung recoil
- E. 4, low lung recoil
- F. 3, low lung recoil
- G. 2, low lung recoil
- H. 1, low lung recoil



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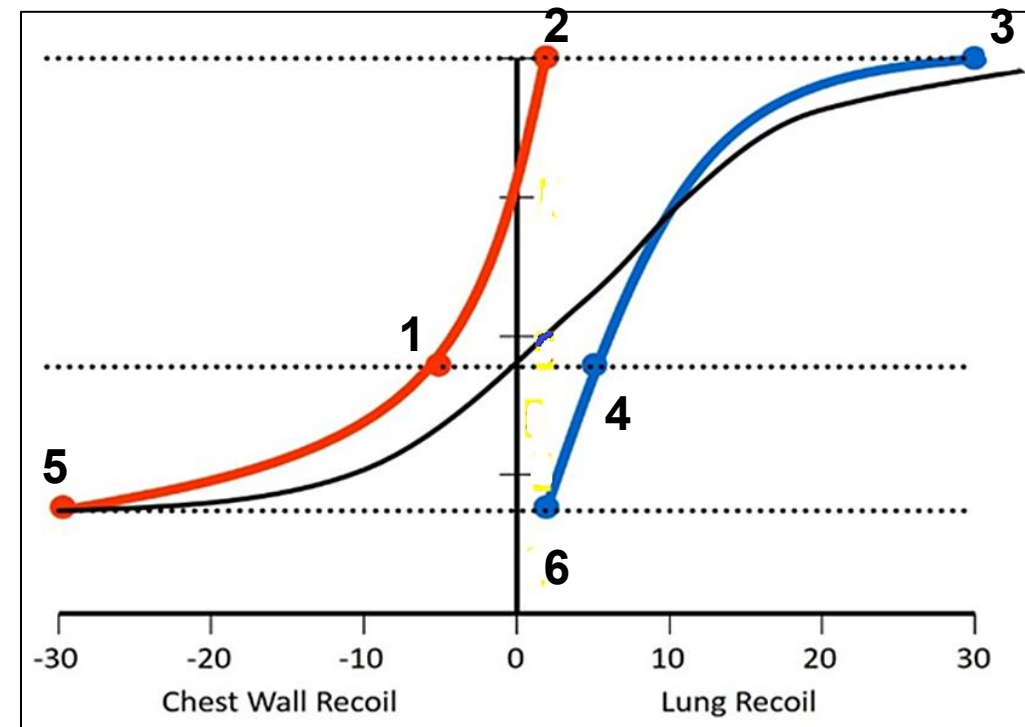


SOM.MK.II.BPM1.3.CPR.2.PHY

S.0207: Explain how lung and chest wall compliance interact to set FRC and how changes in lung or chest wall recoil can change in a patient with obstructive/restrictive disease or with pneumothorax..

3. A 32-year-old woman comes to the physician for a follow-up examination. She had recovered from a lower respiratory infection 3 months ago. Currently she does not have any symptoms. Vital signs are within normal limits. Physical examination shows no abnormalities. The compliance curves of her chest, lungs, and combined (lung+chest) during a respiratory cycle is shown. Which of the following labeled points in the graph best represents the highest chest wall recoil in this patient?

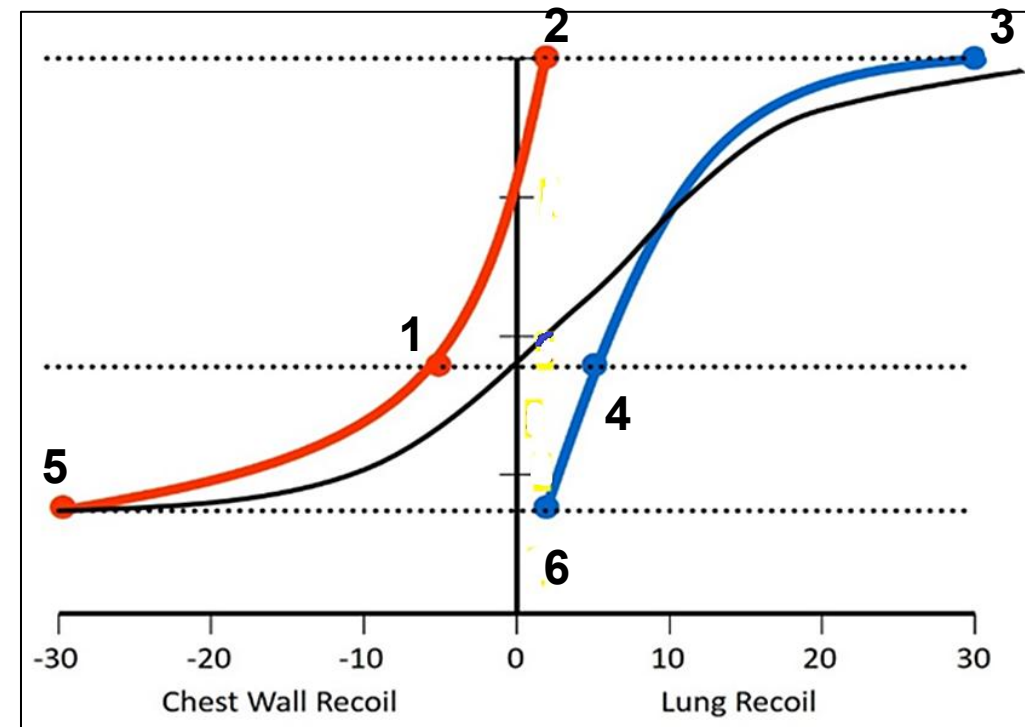
- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6



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- A. 1
- B. 2
- C. 3
- D. 4
- ✓ E. 5
- F. 6

SOM.MK.II.BPM1.3.CPR.2.PHY S.0205 Illustrate and explain the pressure-volume (compliance) curves for the lungs, chest wall, and respiratory system on the same set of axes. Show and explain the significance of the resting positions for each of these three structures.



4. A 59-year-old woman comes to the physician because of a 6-week history of cough and shortness of breath. She has smoked 2 packs of cigarettes daily for 15 years. Her blood pressure is 128/84 mmHg, pulse is 90/min, and respirations are 20/min. Physical examination shows an increased antero-posterior diameter of the chest. Pulmonary function studies show a dead space volume of 200 mL, a tidal volume of 450 mL, and a FEV₁/FVC ratio of 68% (Normal: 80%). Which of the following values best represents alveolar ventilation in this patient?

- A. 3000 mL/min
- B. 4000 mL/min
- C. 45000 mL/min
- D. 5000 mL/min
- E. 7500 mL/min
- F. 9000 mL/min

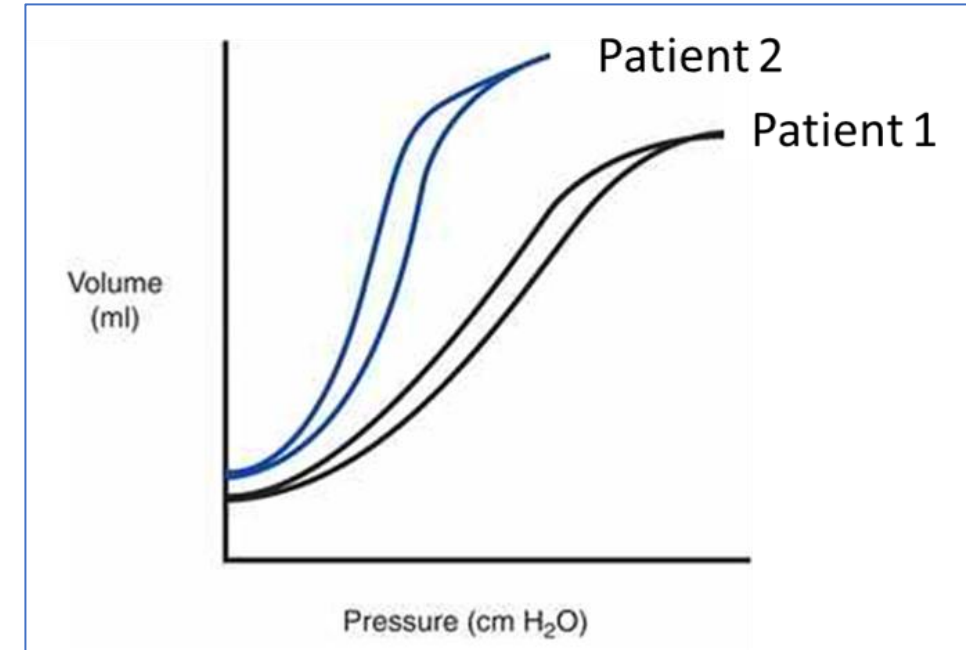
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SOM.MK.II.BPM1.3.CPR.2.PHYS.0237: Describe in quantitative terms the effect of ventilation on PCO₂ according to the alveolar ventilation equation.

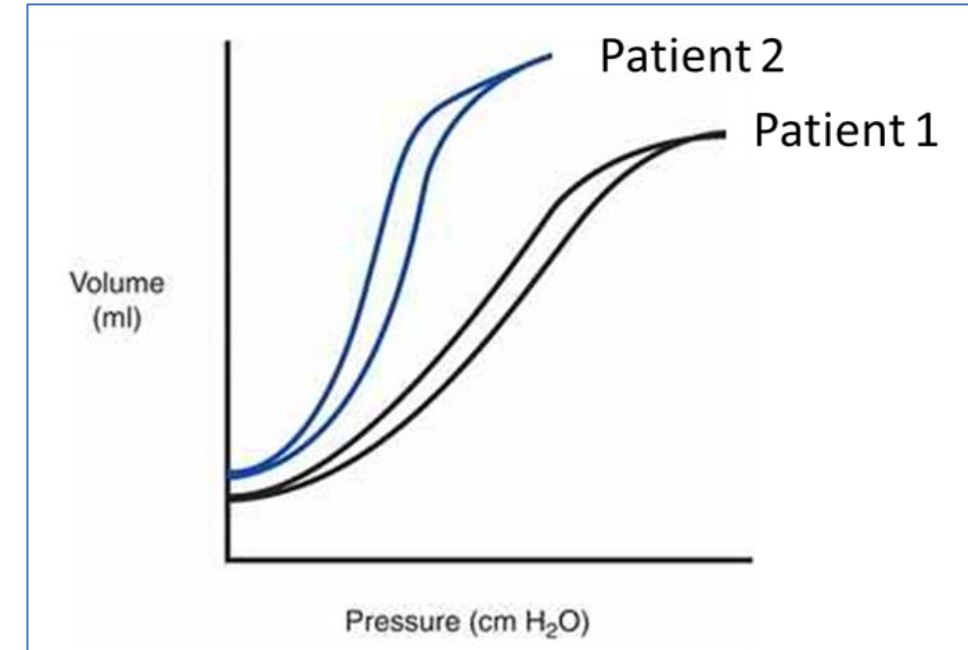
Janine

5. The pressure-volume relationship of the lung of Patient 1 and Patient 2 are shown in the graph. Patient 2 is a chronic smoker and patient 1 is a non-smoker. Which of the following lung changes best explains the position of the pressure-volume relationship of the lung of patient 2 compared to that of patient 1?



- A. Collapse of basal lung segments
- B. Decreased airway diameter
- C. Increased compliance of the lungs
- D. Decreased lung surfactant concentration in the alveolar space
- E. Increased elasticity of the lungs

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- E. Increased elasticity of the lungs

SOM.MK.II.BPM1.3.CPR.2.PHYS.0203:
Compare compliance changes observed with an obstructive or restrictive disease.

6. A 45-year-old man comes to the physician for a follow-up examination. One year ago, he had a Covid-19 infection and recovered completely. Vital signs are within normal limits. Physical examination shows no abnormalities. Results of pulmonary function studies is shown:

Intrapleural pressure measured at the beginning of quiet inspiration = -6 cm H₂O;

Intrapleural pressure measured at the end of quiet inspiration = -12 cm H₂O and

Intrapleural pressure measured at the end of maximal inspiration = -26 cm H₂O.

Which of the following is the best estimate of the transpulmonary pressure at the end of quiet inspiration in this patient?

A. 0 cm H₂O

B. +6 cm H₂O

C. +8 cm H₂O

D. +12 cm H₂O

E. +26 cm H₂O

F. -10 cm H₂O

$$\begin{aligned} P_{TP} &= P_A - P_{IP} \\ &= 0 - (-12) \\ &= +12 \end{aligned}$$

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$$\begin{aligned} P_{TP} &= P_A - P_{IP} \\ &= 0 - (-12) \\ &= +12 \end{aligned}$$

7. A 65-year-old man comes to the physician for a follow-up examination. He has a 20-year history of chronic obstructive lung disease. He has smoked 1 pack of cigarettes daily for 40 years. His vitals are within normal limits. Physical examination shows a barrel-shaped chest and pursed-lip breathing. Pulmonary function studies show increased total lung capacity (TLC) and residual volume (RV). Which of the following mechanisms best explains these changes in lung volumes in this patient?

- A. Decreased lung compliance
- B. Destruction of alveolar septa and reduced lung elastic recoil
- C. Thickened alveolar membranes
- D. Increased lung surface tension
- E. Airway inflammation

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SOM.MK.II.BPM1.3.CPR.2.PHYS.0366 Describe the mechanisms responsible for the changes in the lung volumes and capacities that occur in patients with emphysema and pulmonary fibrosis.

8. A 52-year-old woman comes to the physician because of a 6-month history of dry cough and progressive exertional dyspnea. She does not smoke. Physical examination shows digital clubbing. Chest auscultation shows end-inspiratory crackles. Pulmonary function studies show a reduced TLC, FRC, and RV but a normal FEV₁/FVC ratio. She is diagnosed with a restrictive lung disease. Which of the following mechanisms best explains these findings in this patient?

- A. Increased lung compliance
- B. Loss of alveolar surface area
- C. Decreased lung distensibility
- D. Dynamic airway collapse during expiration
- E. Increased airway resistance

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SOM.MK.II.BPM1.3.CPR.2.PHYS.0366 Describe the mechanisms responsible for the changes in the lung volumes and capacities that occur in patients with emphysema and pulmonary fibrosis.

9. A 55-year-old man comes to the physician for a routine examination. He has smoked 2 packs of cigarettes daily for 20 years. Pulmonary examination shows hyperresonant percussion note, diminished breath sounds with prolonged expiration, and an increased antero-posterior diameter of chest. He is advised pulmonary function studies. During the study, he performs a forced expiration from total lung capacity. The flow initially increases but reduces soon with a prolongation of expiration despite greater effort. This is because of the movement of the equal pressure point (EPP) more distally in the airways than normal. Which of the following mechanisms best explains this shift of the EPP and flow limitation in the patient?

- A. Loss of support in the smaller airways
- B. Increased elastic recoil of the lung tissue
- C. Elevated airway smooth muscle tone
- D. Thickened bronchial wall with mucus hypersecretion
- E. Decreased intrapleural pressure during expiratory effort

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SOM.MK.II.BPM1.3.CPR.2.PHYS.0368 Describe how dynamic compression of airways influences the shape of the effort-independent region of the maximal effort flow-volume curve .

10. A 59-year-old woman comes to the physician for a follow-up examination. She says her shortness of breath has worsened during the last 3 months. She was diagnosed with pulmonary fibrosis 2 years ago. The results of pulmonary function studies using spirometry and helium dilution technique are shown:

Expiratory reserve volume = 0.8 L

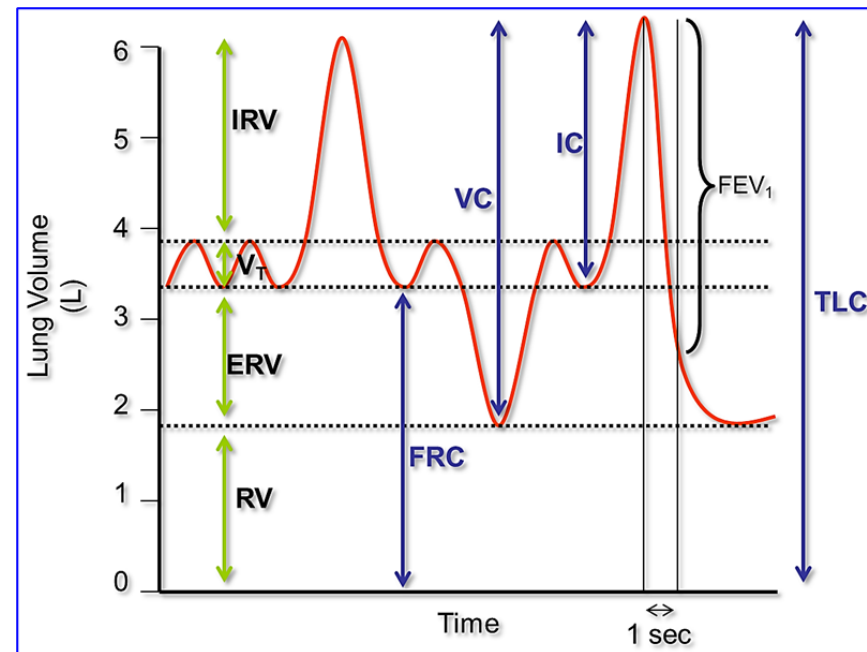
Tidal volume = 0.4 L

Vital capacity = 3.5 L

Functional Residual Capacity = 1.5 L

Which of the following values best represents residual volume (RV) in this patient?

- A. 0.3 L
- B. 0.5 L
- C. 0.7 L
- D. 1.0 L
- E. 1.2 L
- F. 1.8 L



$$\begin{aligned} \text{FRC} &= \text{ERV} + \text{RV} \\ \text{RV} &= \text{FRC} - \text{ERV} \\ &= 1.5 - 0.8 = 0.7. \end{aligned}$$

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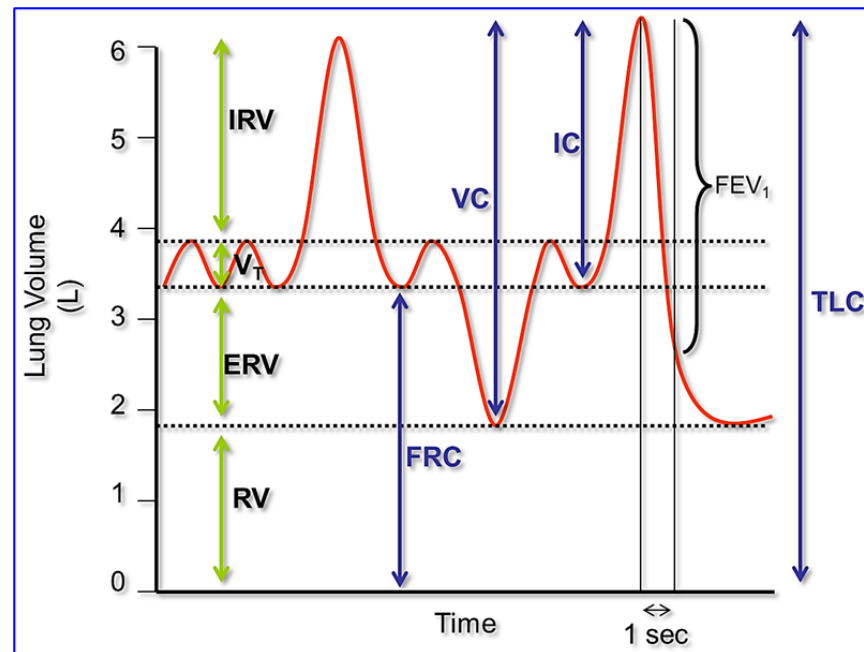
B. 0.5 L

✓ C. 0.7 L

D. 1.0 L

E. 1.2 L

F. 1.8 L



SOM.MK.II.BPM1.3.CPR.2.PHYS.0221:

Draw a normal spirogram, labeling the four lung volumes and four capacities. List the volumes that comprise each of the four capacities. Identify which volume and capacities cannot be measured by spirometry.

$$\text{FRC} = \text{ERV} + \text{RV}$$

$$\text{RV} = \text{FRC} - \text{ERV}$$

$$= 1.5 - 0.8 = 0.7.$$

CPR IMCQ 7

Fall 2025

Biochemistry (3 Questions)

11. A 32-year-old man is brought to the emergency department by his relatives after being found unconscious in his garage. Physical examination shows a confused man with tachypnea. His skin appears pink and flushed. Pulse oximetry shows an oxygen saturation of 99%. Arterial blood gas analysis shows the following: PaO_2 of 95 (normal: 80-100%), SaO_2 of 70% (normal: 95-100%). Which of the following pathophysiologic changes is most likely observed in this patient's hemoglobin?

- A. Contains Fe^{3+} in the porphyrin ring
- B. Stabilized in the “taut” or T state
- C. P50 is increased
- D. More ionic bonds form at the α - β interface
- E. Leftward shift of the oxygen dissociation curve

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SOM.MK.I.BPM1.1.CPR.2.BCHM. 0292; Outline how hemoglobin participates in oxygen and carbon dioxide transport in blood

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0292; Discuss how carbon monoxide (CO) binds to Hb and how this interferes with normal oxygen binding.

12. A 25-year-old man comes to the physician for a follow-up examination. He has relocated from the coast to a city 3000 m (9850 ft) above sea level 2 weeks ago. On the second day, he had headache and nausea, which were resolved after hydration. He says his symptoms have resolved, and he feels better at this visit. Which of the following physiological adaptations is most likely seen in this patient currently when compared to the second day?

- A. Low 2,3-bisphosphoglycerate levels
- B. Lower hemoglobin P50
- C. Lower hemoglobin oxygen affinity
- D. Left shift of the oxygen dissociation curve
- E. PCO_2 higher than reference range
- F. Lower erythropoietin secretion

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- E. PCO_2 higher than reference range
- F. Lower erythropoietin secretion

13. A 25-year-old man comes to the physician because of a 2-day history of fatigue and shortness of breath. He had a common cold a week ago. His temperature is 37.4°C (100°F), pulse is 122/min, blood pressure is 110/70 mm Hg, and respirations are 20/min. Physical examination shows scleral icterus and moderate splenomegaly. Enzyme assay shows 15% activity of pyruvate kinase in red blood cells. Which of the following would most likely be observed in this patient?

- A. Left shift of the oxygen dissociation curve
- B. Increased hemoglobin oxygen affinity
- C. Increased ATP production
- D. Stabilization of the taut state of hemoglobin
- E. Decreased levels of 2,3-bisphosphoglycerate (2,3-BPG)

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- E. Decreased levels of 2,3-bisphosphoglycerate (2,3-BPG)

CPR IMCQ 7

Fall 2025

Genetics (3 Questions)

14. A 3-year-old boy is brought to the physician by his parents for a follow-up examination. Physical examination shows mild hepatosplenomegaly. At a previous visit, laboratory studies showed hemoglobin of 7.5g/dL (normal: 13-17g/dL). Peripheral blood smear shows microcytic anemia. Hemoglobin electrophoresis shows decreased HbA, HbA₂, and HbF, along with a fast-migrating band composed of β -globin tetramers. Genetic testing of the patient and his parents is performed. Which of the following best represents the results in this family?

	Answer Choice	Genotype of Patient	Parent #1 Genotype	Parent #2 Genotype
A.	A	$-/- -/-$	$-/- \alpha/\alpha$ (cis)	$-/- \alpha/\alpha$ (cis)
B.	B	$-/- -/-$	$-/- \alpha/\alpha$ (cis)	$-/\alpha -/\alpha$ (trans)
C.	C	$-/- -/\alpha$	$-/- \alpha/\alpha$ (cis)	$-/\alpha -/\alpha$ (trans)
D.	D	$-/- -/\alpha$	$-/- \alpha/\alpha$ (cis)	$-/- \alpha/\alpha$ (cis)
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D.	D	$-/- -/\alpha$	$-/- \alpha/\alpha$ (cis)	$-/- \alpha/\alpha$ (cis)
E.	E	$-/- -/\alpha$	$-/\alpha -/\alpha$ (trans)	$-/\alpha -/\alpha$ (trans)

15. A 5-year-old boy is brought to the emergency department by his parents because of swelling and pain in his right knee immediately following a minor fall. He has a history of easy bruising and prolonged bleeding after minor injuries. His maternal uncle has a similar condition. Physical examination shows a warm, swollen knee with limited range of motion. Results of laboratory studies are shown. Which of the following genetic mechanisms most commonly explains this patient’s clinical presentation?

Test	Patient Value	Normal Range
Hemoglobin	13.2 g/dL	13.5–17.5 g/dL (men), 12–16 g/dL (women)
Platelets	150,000/mm ³	150,000–400,000/mm ³
Leukocytes (WBC)	8,500/mm ³	4,000–11,000/mm ³
PT	14 sec	11–15 sec
aPTT	112 sec	25–35 sec

- A. Deletion of promoter
- B. Misalignment during homologous recombination
- C. Deletion of exon 26
- D. Deletion of repetitive non-coding regions
- E. Nonsense mutation in exon 22

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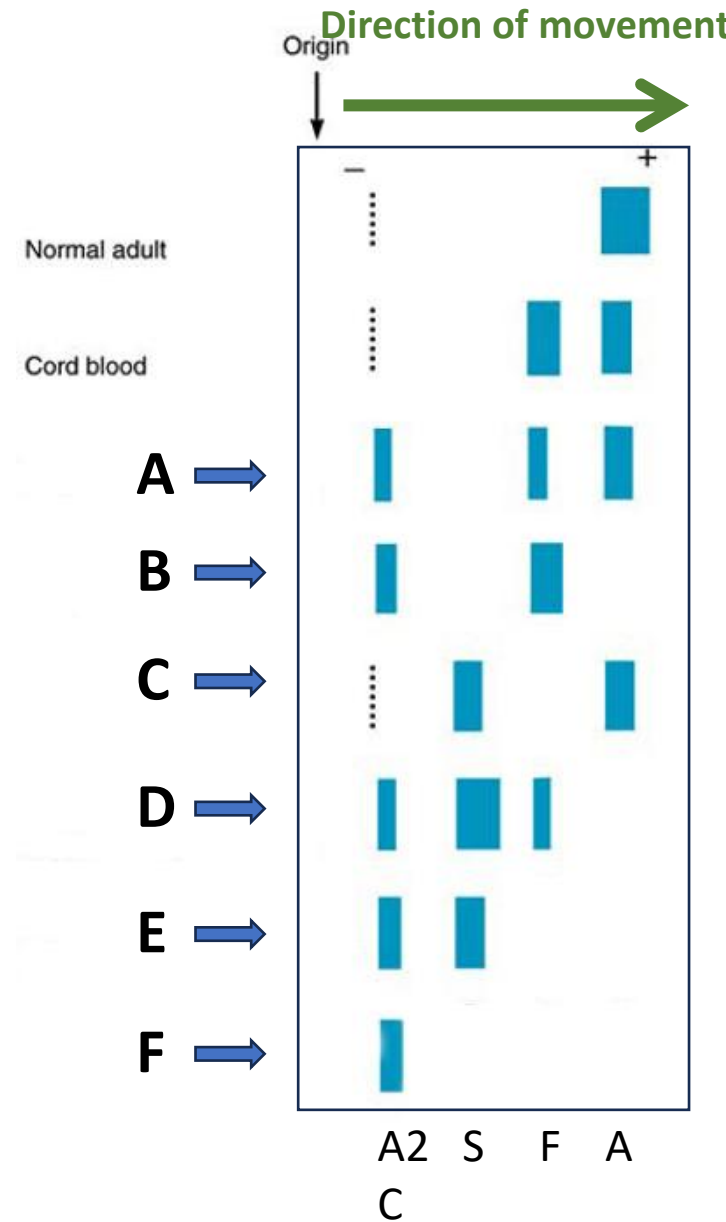
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PT	14 sec	11–15 sec
aPTT	112 sec	25–35 sec

SOM.MK.III.BPM1.1.CPR.3.GNET.0190 Explain the mode of inheritance of hemophilia A and B, the common clinical features and the most common mutation (Intron inversion) in hemophilia A.

- A. Deletion of promoter
- ✓ B. Misalignment during homologous recombination
- C. Deletion of exon 26
- D. Deletion of repetitive non-coding regions
- E. Nonsense mutation in exon 22

16. A 3-year-old boy is brought to the clinic by his parents for evaluation of fatigue and poor growth. Physical examination shows pallor, frontal bossing, malar prominence and hepatosplenomegaly. Laboratory studies show hemoglobin of 7.1 g/dL (normal: 13-17g/dL) and microcytic anemia. Hemoglobin electrophoresis is performed, and the results are shown. Which of the following labeled hemoglobin electrophoresis patterns most likely corresponds to this patient?

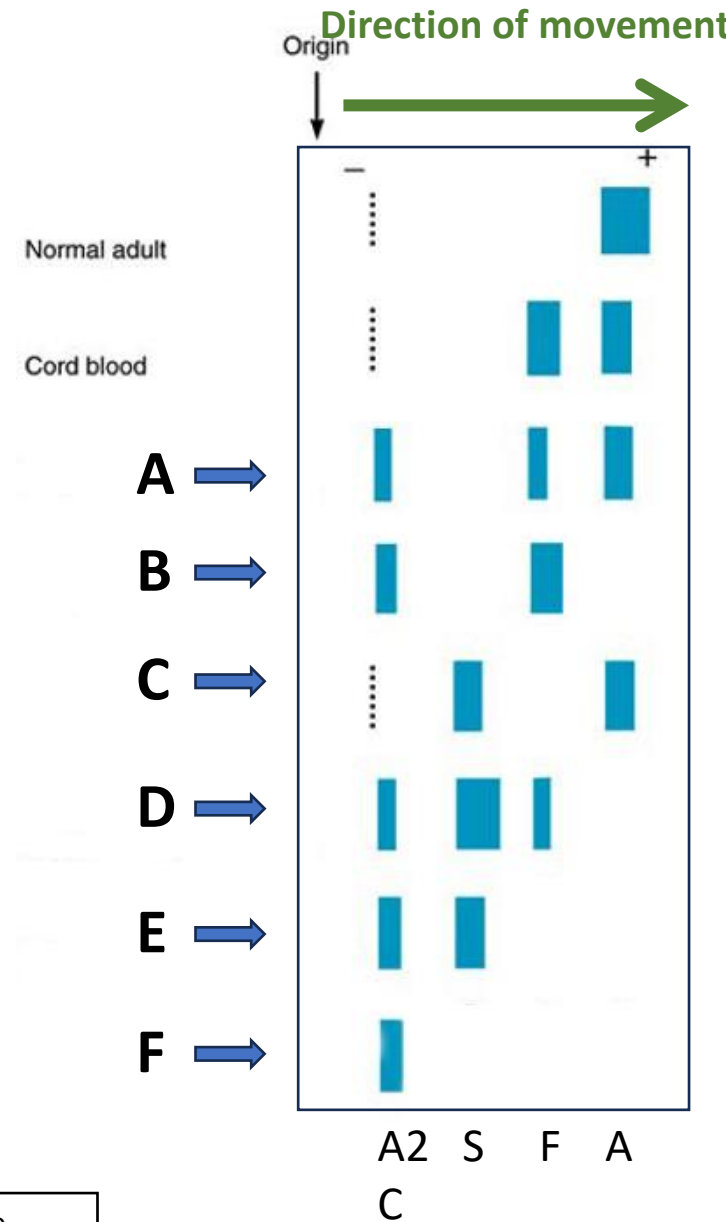
- A. A
- B. B
- C. C
- D. D
- E. E
- F. F



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- A. A
- ✓ B. B
- C. C
- D. D
- E. E
- F. F

SOM.MK.III.BPM1.1.CPR.3.GNET.0182 Explain the effect on the protein due to a single base substitution in HbS and HbC.



ANSWER KEY

- | | |
|------|-------|
| 1. B | 9. A |
| 2. E | 10. C |
| 3. E | 11. E |
| 4. D | 12. C |
| 5. C | 13. D |
| 6. D | 14. C |
| 7. B | 15. B |
| 8. C | 16. B |